

Numerical Time Stepping and Error Evidence

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Notation: Time grid and numerical states

Let $J \subseteq \mathbb{R}$ be an interval, let $X \subseteq \mathbb{R}$, and consider

$$\dot{x} = f(t, x), \quad x(t_0) = x_0,$$

where $f : J \times X \rightarrow \mathbb{R}$. Let $x : I_x \rightarrow X$ denote the exact solution through (t_0, x_0) , where $I_x \subseteq J$ is its interval of definition. Choose a step size $h > 0$ and define

$$t_n = t_0 + nh, \quad n = 0, 1, 2, \dots,$$

whenever $t_n \in I_x$. The symbol $x_n \in X$ denotes the numerical approximation to $x(t_n)$.

Definition: One-step method

For each $h > 0$, let $D_h \subseteq J \times X$ and let $\Psi_h : D_h \rightarrow X$. A **one-step method** computes

$$x_{n+1} = \Psi_h(t_n, x_n)$$

whenever $(t_n, x_n) \in D_h$.

Definition: Local defect

For a one-step method and an exact solution $x(t)$, the **local defect** at step $n + 1$ is the unscaled exact-data error

$$d_{n+1}(h) = x(t_{n+1}) - \Psi_h(t_n, x(t_n)).$$

This quantity is defined whenever $t_{n+1} \in I_x$ and $(t_n, x(t_n)) \in D_h$. Some texts call $d_{n+1}(h)/h$ the local truncation error; here the term **local defect** always means the unscaled quantity above.

Definition: Global error

Let $x(t)$ be the exact solution and let x_n be the numerical state obtained from the same initial value. The **global error** at t_n is

$$e_n = x(t_n) - x_n.$$

Definition: Global convergence order

Fix an initial-value problem whose exact solution exists on $[t_0, T]$, where $T > t_0$. A method has **global order** p on $[t_0, T]$, where $p > 0$, if there are constants $C > 0$ and $h_0 > 0$, independent of h , such that

$$\max_{\{n: t_n \leq T\}} |e_n| \leq Ch^p$$

whenever $0 < h < h_0$. This definition also requires the numerical states in the displayed maximum to exist.

Idea: Euler's method is left-endpoint quadrature

The exact solution satisfies

$$x(t_{n+1}) = x(t_n) + \int_{t_n}^{t_{n+1}} f(t, x(t)) dt.$$

Replacing the integrand by its left-endpoint value gives

$$x_{n+1} = x_n + hf(t_n, x_n).$$

Question: What error orders appear for Euler's method?

Apply Euler's method to

$$\dot{x} = x, \quad x(0) = x_0 \neq 0.$$

Determine the order of the unscaled local defect and the order of the global error after a fixed time $T > 0$.

Solution. All estimates below are as $h \downarrow 0$ with T fixed. The local-defect estimates are uniform over grid points in $[0, T]$. Starting from the exact value $x(t_n)$,

$$x(t_n + h) = e^h x(t_n), \quad \Psi_h(t_n, x(t_n)) = (1 + h)x(t_n).$$

Thus the one-step defect is

$$e^h x(t_n) - (1 + h)x(t_n) = x(t_n) \left(\frac{h^2}{2} + O(h^3) \right),$$

which is $O(h^2)$. At any grid point $t_n = nh \leq T$, Euler gives

$$x_n = (1 + h)^n x_0.$$

Since

$$\log(1 + h) = h - \frac{h^2}{2} + O(h^3),$$

Question: What error orders appear for Euler's method? (continued)

we obtain

$$(1+h)^n = \exp\left(t_n - \frac{t_n}{2}h + O(h^2)\right).$$

Subtracting from $x(t_n) = x_0 e^{t_n}$ gives

$$e_n = x_0 e^{t_n} \left(\frac{t_n}{2}h + O(h^2) \right).$$

These estimates are uniform for $0 \leq t_n \leq T$. Thus the local defect is $O(h^2)$ and the global error on $[0, T]$ is $O(h)$.

Definition: Improved Euler method

The **improved Euler method** first predicts

$$\tilde{x}_{n+1} = x_n + hf(t_n, x_n)$$

and then accepts the average-slope update

$$x_{n+1} = x_n + \frac{h}{2} [f(t_n, x_n) + f(t_n + h, \tilde{x}_{n+1})].$$

For an autonomous equation, the explicit time arguments are omitted.

Question: What error orders appear for improved Euler?

Apply improved Euler to

$$\dot{x} = x, \quad x(0) = x_0 \neq 0.$$

Determine the order of the unscaled local defect and the order of the global error after a fixed time $T > 0$.

Solution. All estimates below are as $h \downarrow 0$ with T fixed. The local-defect estimates are uniform over grid points in $[0, T]$. The predictor is

$$\tilde{x}_{n+1} = (1+h)x_n.$$

The accepted update is

$$x_{n+1} = x_n + \frac{h}{2} (x_n + \tilde{x}_{n+1}) = \left(1 + h + \frac{h^2}{2}\right) x_n.$$

Since

$$e^h - \left(1 + h + \frac{h^2}{2}\right) = \frac{h^3}{6} + O(h^4),$$

the local defect is $O(h^3)$. Moreover,

$$\log\left(1 + h + \frac{h^2}{2}\right) = h - \frac{h^3}{6} + O(h^4),$$

so at any grid point $t_n = nh \leq T$,

$$\left(1 + h + \frac{h^2}{2}\right)^n = \exp\left(t_n - \frac{t_n}{6}h^2 + O(h^3)\right).$$

Question: What error orders appear for improved Euler? (continued)

Subtracting from $x(t_n) = x_0 e^{t_n}$ gives

$$e_n = x_0 e^{t_n} \left(\frac{t_n}{6} h^2 + O(h^3) \right).$$

These estimates are uniform for $0 \leq t_n \leq T$. Thus the local defect is $O(h^3)$ and the global error on $[0, T]$ is $O(h^2)$.

Definition: Classical Runge–Kutta method

For $\dot{x} = f(t, x)$, define

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f(t_n + h/2, x_n + hk_1/2), \\ k_3 &= f(t_n + h/2, x_n + hk_2/2), \\ k_4 &= f(t_n + h, x_n + hk_3). \end{aligned}$$

The **classical fourth-order Runge–Kutta method** is

$$x_{n+1} = x_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Theorem: Order of the classical Runge–Kutta method

Let $D \subseteq \mathbb{R}^2$ be open, let $f \in C^4(D)$, and suppose the graph of the exact solution of $\dot{x} = f(t, x)$ on $[t_0, T]$ lies in a compact subset of D . If f is Lipschitz in x on a neighborhood of that graph, then, for all sufficiently small $h > 0$, its numerical states exist through time T and there are constants $C_1, C_2 > 0$ such that

$$\max_{\{n: t_{n+1} \leq T\}} |d_{n+1}(h)| \leq C_1 h^5, \quad \max_{\{n: t_n \leq T\}} |e_n| \leq C_2 h^4.$$

Question: How do three methods compare on linear decay?

For

$$\dot{x} = -2x, \quad x(0) = 1,$$

use $h = 1/2$ to approximate $x(1)$ by Euler, improved Euler, and classical Runge–Kutta.

Solution. The exact value is

$$x(1) = e^{-2} \approx 0.135335.$$

Euler gives

$$x_1 = 1 + \frac{1}{2}(-2) = 0, \quad x_2 = 0.$$

For improved Euler,

$$\tilde{x}_1 = 1 + \frac{1}{2}(-2) = 0, \quad x_1 = 1 + \frac{1}{4}(-2 + 0) = \frac{1}{2},$$

and

$$\tilde{x}_2 = \frac{1}{2} + \frac{1}{2}(-1) = 0, \quad x_2 = \frac{1}{2} + \frac{1}{4}(-1 + 0) = \frac{1}{4}.$$

Question: How do three methods compare on linear decay? (continued)

For the first Runge–Kutta step,

$$(k_1, k_2, k_3, k_4) = \left(-2, -1, -\frac{3}{2}, -\frac{1}{2}\right),$$

so

$$x_1 = 1 + \frac{1/2}{6} \left(-2 - 2 - 3 - \frac{1}{2}\right) = \frac{3}{8}.$$

For the second step,

$$(k_1, k_2, k_3, k_4) = \left(-\frac{3}{4}, -\frac{3}{8}, -\frac{9}{16}, -\frac{3}{16}\right),$$

and

$$x_2 = \frac{3}{8} + \frac{1/2}{6} \left(-\frac{3}{4} - \frac{3}{4} - \frac{9}{8} - \frac{3}{16}\right) = \frac{9}{64} = 0.140625.$$

Visualization: Slope field for the logistic equation

For

$$\dot{x} = x(1 - x),$$

the slope depends only on x , so it is constant along every horizontal line in the (t, x) plane:

x	$\dot{x}$	time-graph direction
$x < 0$	$-$	
$x = 0$	0	horizontal
$0 < x < 1$	$+$	
$x = 1$	0	horizontal
$x > 1$	$-$	

The equilibrium solution graphs are $x(t) = 0$ and $x(t) = 1$.

Theorem: Step-halving signature

Let $Q \in \mathbb{R}$, and suppose $Q_h \in \mathbb{R}$ is defined for every sufficiently small $h > 0$. Suppose, as $h \downarrow 0$,

$$Q_h = Q + Ch^p + o(h^p), \quad C \in \mathbb{R} \setminus \{0\}, \quad p > 0.$$

Then

$$\frac{Q_h - Q_{h/2}}{Q_{h/2} - Q_{h/4}} \rightarrow 2^p,$$

and therefore

$$p_{\text{obs}} = \log_2 \left| \frac{Q_h - Q_{h/2}}{Q_{h/2} - Q_{h/4}} \right| \rightarrow p.$$

Proof. The expansion gives

$$Q_h - Q_{h/2} = Ch^p(1 - 2^{-p}) + o(h^p)$$

and

$$Q_{h/2} - Q_{h/4} = Ch^p 2^{-p}(1 - 2^{-p}) + o(h^p).$$

The denominator has a nonzero leading coefficient because $C \neq 0$ and $p > 0$, so it is nonzero for sufficiently small h . Division gives the limit 2^p ; continuity of $\log_2$ then gives $p_{\text{obs}} \rightarrow p$. $\square$

Corollary: Richardson extrapolation

Under the hypotheses of the step-halving theorem,

$$Q = Q_{h/2} + \frac{Q_{h/2} - Q_h}{2^p - 1} + o(h^p).$$

Proof. Multiplying the finer approximation by 2^p cancels the leading errors:

$$\begin{aligned} 2^p Q_{h/2} - Q_h &= (2^p - 1)Q + (2^p C(h/2)^p - Ch^p) + o(h^p) \\ &= (2^p - 1)Q + o(h^p). \end{aligned}$$

Divide by $2^p - 1$ and rewrite the left side to obtain the stated formula. □

Question: What order is visible in an error table?

For $h = 0.2, 0.1, 0.05$, suppose the absolute errors are

$$1.6 \times 10^{-3}, \quad 4.1 \times 10^{-4}, \quad 1.0 \times 10^{-4}.$$

Estimate the observed order.

Solution. If an error has the leading form Ch^p , then halving h divides the error by approximately 2^p . Therefore the base-2 logarithm of each consecutive error ratio estimates p . The two refinement estimates are

$$\log_2 \left(\frac{1.6 \times 10^{-3}}{4.1 \times 10^{-4}} \right) \approx 1.96$$

and

$$\log_2 \left(\frac{4.1 \times 10^{-4}}{1.0 \times 10^{-4}} \right) \approx 2.04.$$

The data are consistent with second-order convergence.